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CS 621 DATA ANALYTICS

Text Processing

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BASICS

- Text provided in written form. Considered as unstructured data.
- Involves Natural Language Processing.
- Uses third party text/language processing libraries.
- NLTK – Natural Language Toolkit (www.nltk.org)
- Python string module

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STRING CONSTANTS

- import string
- print(string.ascii_letters)
- print(string.ascii_uppercase)
- print(string.ascii_lowercase)
- print(string.digits)
- print(string.punctuation)
- print(string.whitespace)

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USEFUL STRING METHODS

- Use these to normalize words:
  - str.lower(), str.upper(), str.capitalize()
  - str.title()
- Use these to examine strings:
  - str.isalnum()
  - str.isalpha()
  - str.isdigit()
  - str.islower()
  - str.isupper()
  - str.isspace()

See class samples for demonstration.

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TEXT INTO SENTENCES

```
>>> para = "Hello there! It is nice to see you again. Do you have any questions?"

>>> from nltk.tokenize import sent_tokenize

>>> sent_tokenize(para)

['Hello there!', 'It is nice to see you again.', 'Do you have any questions?']
```

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SENTENCES INTO WORDS

```
>>> from nltk.tokenize import word_tokenize

>>> word_tokenize('It is nice to see you again.')

['It', 'is', 'nice', 'to', 'see', 'you', 'again', '.']
```

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REMOVE PUNCTUATION MARKS

- string.punctuation

!"#$%&\'()*+,-./:;<=>?@[\\]^_`{|}~

```
>>> s = 'It is nice to see you again.'
>>> print(s.translate(str.maketrans('','', string.punctuation)))
```

It is nice to see you again

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STOP WORDS REMOVAL

- A stop word is a commonly used word such as “the”, “a”, “you”, “he”, etc. These words do not add much meaning to a sentence.

```
>>> from nltk.corpus import stopwords
>>> english_stops = set(stopwords.words('english'))
>>> words_saved = []
    for w in words_list:
            if w not in english_stops:
                    words_saved.append(w)
```

- You may want to construct your own stop words list for your own application domain.

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STEMMING

- Stemming is a technique to remove affixes from a word, ending up with the stem.
- Cooking -> Cook
- Porter and Lancaster algorithms

```
>>> porter = nltk.PorterStemmer()
    porter.stem('helped")
```

>>>help is returned.

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TWO STEMMERS

```
>>> raw = """DENNIS: Listen, strange women lying in ponds distributing swords ... is no basis for a system of government. Supreme executive power derives from ... a mandate from the masses, not from some farcical aquatic ceremony."""  # example from the nltk book.
>>> tokens = word_tokenize(raw)

>>> porter = nltk.PorterStemmer()

>>> lancaster = nltk.LancasterStemmer()

>>> [porter.stem(t) for t in tokens]

>>> [lancaster.stem(t) for t in tokens]
```

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LEMMATIZATION

- Similar to stemming; to reduce words to their base form.
- A lemma is a root word.
- The WordNet lemmatizer only removes affixes if the resulting word is in its dictionary

```
>>>
    nltk.WordNetLemmatizer().lemmatize('helps')
    help
    nltk.WordNetLemmatizer().lemmatize('helped')
    helped
    nltk.WordNetLemmatizer().lemmatize('helping')
    helping
```

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A SHORT SUMMARY

- Load text data from a local file or a data source on Internet.
- Save the text in appropriate data structure(s).
- Perform the following tasks:
  - remove punctuation marks in a sentence.
  - retrieve each word in each sentence.
  - convert all characters in each word in lowercase or uppercase.
  - remove stop words
  - sort the words left if needed
  - perform isalpha(), isalnum(), isdigit() if needed.
  - output the cleaned words into a text file or a csv file



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